J. Earth Syst. Sci. (2024)133 5
https://doi.org/10.1007/s12040-023-02210-1

© Indian Academy of Sciences

Application of XGBoost model for early prediction of earthquake magnitude from waveform data

Anushka Joshi[1,*], Chalavadi Vishnu[2], C Krishna Mohan[2] and Balasubramanian Raman[1]

[1]*Department of Computer Science and Engineering, Indian Institute of Technology, Roorkee 247 667, India.*
[2]*Department of Computer Science and Engineering, Indian Institute of Technology, Hyderabad 502 285, India.*
[*]*Corresponding author. e-mail: anushka_j@cs.iitr.ac.in anushkajoshi1374@gmail.com*

MS received 11 April 2023; revised 24 August 2023; accepted 28 August 2023

In this paper, a scalable end-to-end tree boosting system called XGBoost has been applied for predicting the magnitude of an earthquake from the early part of earthquake waveform data. This model uses the features extracted from the early P wave phase of the records as an input. The model's effectiveness has been verified by using data on earthquakes occurring in the Eurasian plate of Japan Islands from 1996 to 2021. Feature engineering has given 29 new features identified from the early P wave phase of the record, which show a high correlation with the magnitude of an earthquake. The comparison of predicted and actual magnitude shows that a trained XGboost model, which uses a single input record for magnitude prediction, gives an average prediction error of 0.004 ± 0.57 for earthquakes in the test dataset. In contrast, the average prediction error of -1.1 ± 0.80 and -0.65 ± 0.69 has been obtained for the magnitude estimated from conventional τ_c and P_d methods using the same test dataset. It is further seen that the average predicted magnitude of a single earthquake of magnitude 4.5 and 6.1 (M_{JMA}) obtained by using multiple nearfield records using XGBoost model is 4.58 ± 0.33 and 6.32 ± 0.29, which is close to the actual magnitude of the earthquake. The results presented in this paper clearly show that the structured data can be effectively used by complex machine learning or deep learning models to predict earthquake magnitude from single or multiple records.

Keywords. Earthquake; early warning; magnitude; machine learning; feature.

1. Introduction

Earthquake early-warning or prediction is among recently developed fields in seismology where the size and location of earthquakes are detected on the basis of early P phases in the record. Earthquake early-warning networks installed in various countries can be broadly divided into two categories, i.e., onsite and regional warning systems. Onsite warning systems are operational in Japan (Nakamura 1989), California (Allen and Kanamori 2003), and Romania (Wenzel *et al.* 1999; Böse *et al.* 2007). These systems make use of predominant periods or the amplitudes of the first few seconds of P waves at single and multiple stations to evaluate the magnitude of ground shaking (Wu *et al.* 2007; Olivieri *et al.* 2008; Shieh *et al.* 2008; Böse *et al.* 2009; Brown *et al.* 2009; Kuyuk and Allen 2013; Picozzi *et al.* 2015; Mousavi and Beroza 2020; Zhang *et al.* 2021; Zhu *et al.* 2021, 2022).

Published online: 22 December 2023

Regional warning systems are operational in Taiwan (Wu *et al.* 2014), Mexico (Cuéllar *et al.* 2014), Japan (Hoshiba 2014), India (Chamoli *et al.* 2021), and the US (Given *et al.* 2014). These systems use records from many stations and upgrade the location of epicentre and magnitude as energy from the earthquake source reaches various stations from the hypocenter. The regional warning systems use arrival times and amplitudes of P waves at multiple stations to estimate earthquake locations and magnitudes for reliable and complete warning information (Kanamori 2005; Allen *et al.* 2009; Cua *et al.* 2009; Picozzi *et al.* 2015). The algorithms used in regional warning systems comprise many steps, including phase detection, phase picking, phase association, earthquake location, and magnitude estimation (Satriano *et al.* 2011; Grigoli *et al.* 2018; Zhang *et al.* 2019). It has been seen that these steps may affect the estimation of the source parameter and require more time to receive and analyse earthquake signals at multiple stations (Zhang *et al.* 2021). In regional warning systems, the early warning cannot be issued until sufficient close stations are triggered.

Both the regional and onsite warning systems are based on parameters picked from the early parts of the P phase, i.e., the peak amplitude of displacement (Wu and Li 2006), predominant period (Nakamura 1988), and characteristic period (Wu and Kanamori 2005). However, it is always the epicentral region that is more likely to be prone to earthquake risk. Hence, it is challenging to correctly predict the magnitude of an earthquake from a few seconds of the early P phase that too from a few nearfield records. Zhang *et al.* (2021) have demonstrated that the seismic waveform contains more information than merely a few parameters that can be transformed into an effective warning system by properly utilising machine learning tools. Machine learning and deep learning approaches provide a suitable platform to analyse the nonlinear dependency of various waveform features directly to the size of an earthquake. Recent studies by Mousavi and Beroza (2020), Perol *et al.* (2018), Reddy and Nair (2013), and Ochoa *et al.* (2017) have demonstrated that machine learning and deep learning tools have excellent potential for use in magnitude estimation in EEW systems (Zhu *et al.* 2021). Among the machine learning methods used in practice, extreme gradient tree boosting is one technique that has many applications (Friedman 2001). A scalable machine learning model named XGBoost based on gradient tree boosting has been applied in several fields, including hazard risk prediction (Chen and Guestrin 2016). XGBoost exploits out-of-core computation and enables data scientists to process hundreds of millions of examples on a desktop. In the present work, the applicability of XGBoost model has been explored for predicting earthquake magnitude in earthquake early warning systems using various extracted features from P phase of the record at single or multiple stations.

2. Data

Seismic activity in Japan is mainly due to the collision of Eurasian, Philippine Sea, North American, and Pacific plates with each other. In this region, the Pacific Plate subducts beneath the North American Plate at a rate of 8 cm/year while the Philippine Sea Plate subducts beneath the North American and Eurasian plates at an approximately 4 and 4–7 cm/year, respectively (Satake 2015). The worldwide distribution of earthquakes clearly shows that the source mechanism of earthquakes is directly related to the nature of plate boundaries with which it is associated. In the present work, only earthquakes occurring within the Eurasian Plate and its periphery have been selected for training and testing of the developed earthquake magnitude prediction model. Kyoshin Network (K-NET) is a network of strong-motion seismographs installed at approximately 1000 locations in Japan. The earthquakes recorded by stations of K-NET lying between 30°–36°N and 128.5°–139°E have been considered in this work. The dynamic range of the accelerograph used in the K-NET is 114 dB or more (Aoi *et al.* 2020). The seismographs utilised in the study have been manufactured by OYO Corp. since 2012 (Aoi *et al.* 2020). The power spectrum density (PSD) plots are the standard method used for quantification of seismic background noise in a station. PSD are compared with New Low Noise Model (NLNM) and New High Noise Model (NHNM) defined by Peterson (1993). The PSD plots at some of the stations that are utilised in the present work are shown in figure 1. Figure 1 suggests that the noise recorded by all the sites eventually falls under NHNM and within NLNM limits defined by Peterson model (1993).

Figure 2 shows the location of the earthquakes that have occurred in this region from 1996 to 2021. In this work, the data have been obtained

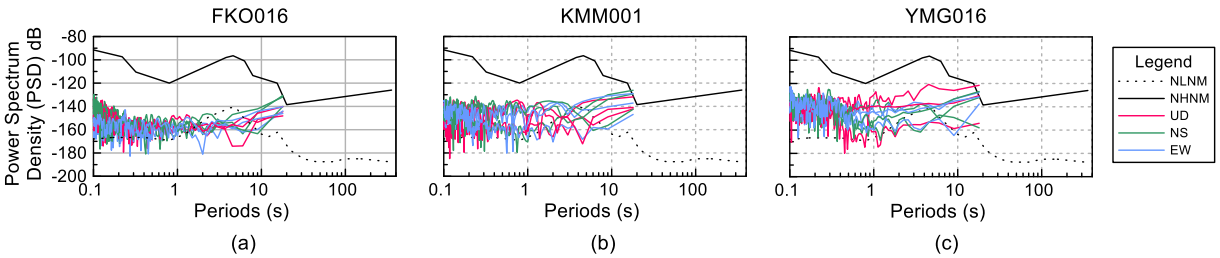


Figure 1. Power spectral density plot of background seismic station noise when there is no earthquake event. The plot is shown with reference of new-low noise model (NLNM) and new-high noise model (NHNM) taken from Peterson (1993). The plots are created using the noise from three components (NS, EW, and UD) of accelerogram of the selected sites named as **(a)** Ohmuta, (**b**) Oguni, and **(c)** Iwakuni.

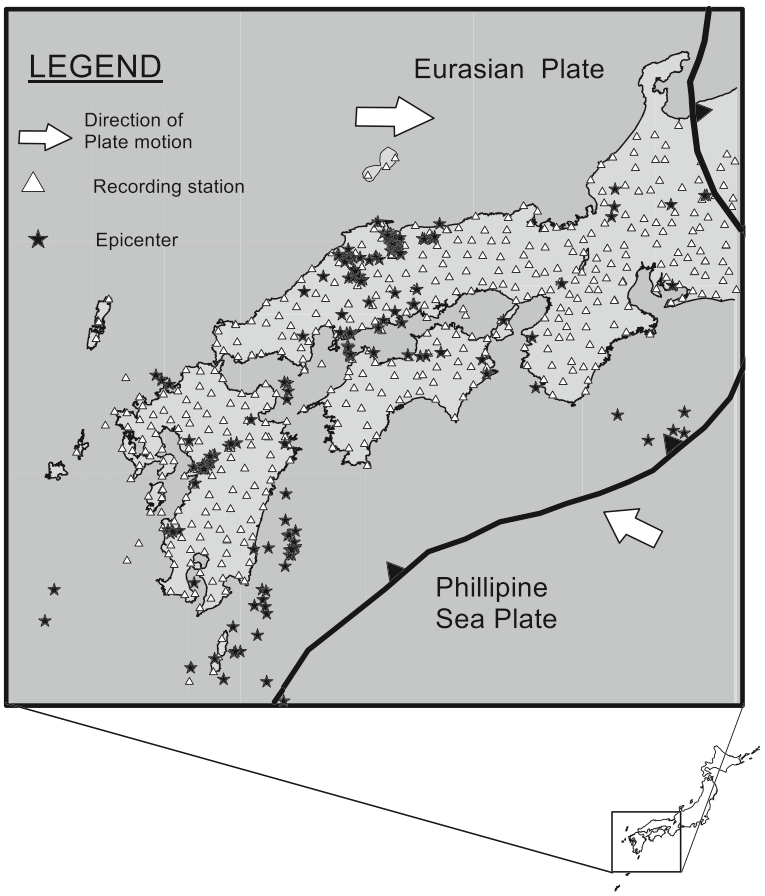


Figure 2. Location of recorded selected stations and epicentre of earthquakes which are used in present work. The station location is shown by a white-filled triangle, while the black star shows the epicenter of the earthquake. The tectonics showing plate boundary and direction of plate motion has been taken after Takla *et al.* (2013).

from the K-NET stations of the National Research Institute for Earth Science and Disaster Prevention (NIED) in Japan (Aoi *et al.* 2011). The location of stations that have recorded this data is shown in figure 2. The magnitude utilised for testing the model is represented in the Japan Meteorological Agency (JMA) scale. The JMA seismic intensity scale has been widely used in Japan since 1996 for identifying structural damage, crisis management preparation, and identifying affected areas due to an earthquake (Yamazaki 1996; Yamazaki *et al.* 1998). The JMA scale is based on instrumental measurements that consider amplitude, frequency, and duration of shaking (Yamamoto *et al.* 2008). The vertical component of acceleration record with a sampling frequency of 100 Hz has been used. The significance of using vertical component for studying damaging effects on structures have been studied by various researchers (Papazoglou and Elnashai 1996; Elgamal and He 2004). The vertical

component has been utilised for the analysis since it has the theoretical advantage that it represents the vertical component of Rayleigh waves alone (Båth 1952). The horizontal component, on the other hand, represents the combination of Rayleigh and Love waves. The practical advantage of utilising the vertical component is that only one record needs to be measured, whereas two records (NS and EW) need to be measured for the horizontal component (Båth 1952).

Strong motion records are always affected by noise to some degree and therefore, data processing is applied to the recorded data (Boore and Bommer 2005). The machine learning application presented in this work utilises several parameters that are extracted directly from strong motion records. The records have been corrected for instrumental scaling and baseline correction. Filtering is the most effective and least subjecting tool for reducing noise in the strong motion record (Boore and Bommer 2005). In order to remove low-frequency noises from the records, the records have been passed through a bandpass filter in the range of 1.0–45.0 Hz. A code in Python named FETS (feature extraction of time series) has been developed to process all strong motion records and extract features from the early P phase of the record for the training of scalable machine learning models.

The EEW systems installed in various parts of the world have been designed to record and process the recorded P phase. However, due to the triggering threshold mechanism, the strong motion data acquired from the K-NET site consists of many records triggered only after the arrival of the S phase. Such records which are devoid of the P phase may pose a threat to the reliability of the present approach based on the features extracted strictly from the P phase of records. In order to remove such records where no clear P phase is present, only those records that contain an appreciable amount of pre-event signals with zero mean amplitude have been considered. Such records have been identified by having a zero mean value up to 0.01 sec before the onset of P phase. The P phase is automatically picked by using the algorithm based on the ratio of short-term with long-term averaging (STA/LTA) given by Allen (1978) with a threshold ratio of 2.0 used in this paper. Further, it is seen that for stations lying very close to the epicentre, the P and S phases overlapped with each other, thereby giving false values of features from early P phases of the record. Hence in the present work, only stations that are lying more than 50 km hypocentral distance have been considered. As hypocentral distance increases, the difference in the arrival of P and S phases is visible in the records, and hence, chances of overlapping of two phases decrease. However, considering the requirement of early warning from stations close to the epicentre, we have restricted the use of data from stations up to 150 km hypocentral distance. It is seen that out of 4600 downloaded records in this area, a total of 2960 records from earthquakes within $3.0 \leq M_{JMA} \leq 7.4$ satisfy this criterion. The entire database has been divided into training and testing datasets with 2360 and 591 records. The testing and training datasets are independent of each other to avoid any biases in the training of the earthquake magnitude prediction model. Figure 3 shows the details of the database of records used for training and testing of the model. It is seen from figure 3 that in the database, small magnitude events are mainly within 100 km epicentre distance, while it increases for magnitude 5.2 (M_{JMA}) onwards. This is mainly due to the less amount of energy released for small magnitude events, which decay directly with distances. The database also shows the scarcity of records of high-magnitude

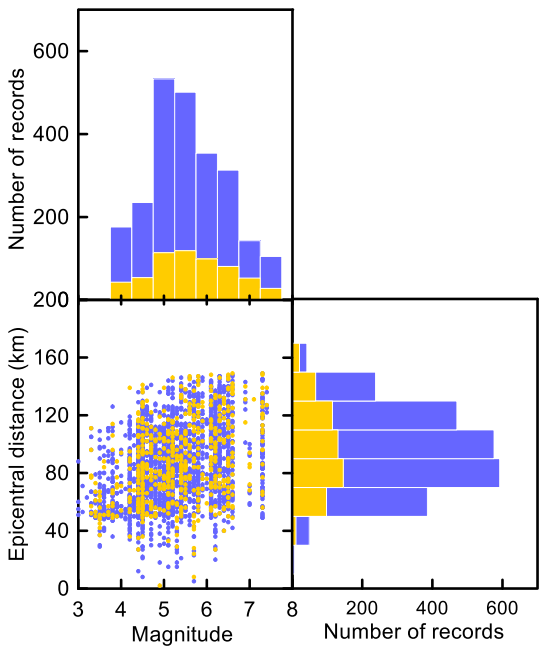


Figure 3. Distribution of records used in the training and testing of the algorithm with respect to the magnitude and epicentral distance (in km). The magnitude interval is 0.5 in the histogram plot for the distribution of the number of earthquakes in the different magnitudes. The epicentral interval in the histogram plot for distribution of records with respect to epicentral distance is 20 km. The training data is shown in blue, while the testing data is shown in yellow.

events mainly because of the low frequency of high-magnitude events.

3. Methodology

New developments in the field of machine learning and deep learning tools have given rise to the possibility of predicting earthquakes to a level that can be utilised for early warning purposes (Alves 2006; Alexandridis *et al.* 2014; Rouet-Leduc *et al.* 2017; Zhu *et al.* 2021). The problem of predicting the magnitude of an earthquake can be potentially solved by combining good data with machine and deep learning tools. The machine learning tool utilised in this paper falls under the category of the supervised learning method. Supervised learning first teaches a model to map input data to output data. After its training, the model is further used to make predictions on test data, which is not utilised for training purposes (Johnson *et al.* 2021). A good dataset is essential for the dependent outcome of machine learning methods.

In this paper, XGBoost, a scalable machine-learning model for tree boosting, has been utilised for EEW. XGBoost stands for Extreme Gradient Boosting and is an ensemble machine learning model. The system is available as an open-source package (Bekkerman *et al.* 2011). The impact of the system has been widely recognised in a number of machine learning and data mining challenges (Chen and Guestrin 2016). The purpose of tree boosting in the XGBoost model is to combine a series of weak classifiers to build a strong classifier (Chen *et al.* 2019). In the present work, the dataset consists of 29 extracted features x_i, $i = 1, \ldots, 29$ from the acceleration record and magnitude y_i corresponding to each record. Assuming a dataset is $D = \{(x_i, y_i) : i = 1, \ldots, n,\ x_i \in R, m, y_i \in R\}$ for n samples with m features, the relation between actual y_i and predicted $\widehat{y}_i$ can be given by the following equation (Chen *et al.* 2019):

$$\widehat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in F \tag{1}$$

where f_k represents an independent regression tree and $f_k(x_i)$ denotes the prediction score given by the kth tree to the ith sample. The objective function for XGBoost model given by Chen *et al.* (2019) is:

$$Obj^{(i)} = \sum_{i=1}^{n} l(y_i, \widehat{y}_i) + \Omega(f_k). \tag{2}$$

In the above equation, l denotes the training loss function, which is a root mean square error in the present work and Ω represents the term used to avoid overfitting. The value of Ω is given as:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda ||w||^2. \tag{3}$$

In the above equation, γ and λ are the degrees of regularisation. T and w are the numbers of leaves and their scores, respectively. The set of functions used in the regression tree model has been learned by minimising the objective function. In the gradient tree boosting method, a new function f_k has been added to minimise the following objective function:

$$L^{(t)} = l\left(y_i + \widehat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t). \tag{4}$$

In this expression, $\widehat{y}_i^{(t)}$ is the prediction of the ith instance at tth iteration. Function f_t has been added to improve our model. It is seen from the objective function of XGBoost method that it is a function of functions. The XGBOOST model is a sum of tree learners, which minimises the logarithm of loss objective function and score at leaves. The score at leaves are actually the weight having a meaning as a sum across all the tree of the model and are always adjusted in order to minimise the loss. The hyperparameters of the XGBoost model are tuned and the model is trained using a training dataset. Finally, the testing of trained model has been done by predicting magnitude from the records in the test datasets. The entire process has been summarised in the form of a flow graph shown in figure 4.

In the present work, the XGBoost model has been compared with other prominent machine learning models like Support Vector Machine (Radial basis function kernel), Lasso Regressor, and Decision Tree Regressor. The prediction error, which defines the sum of the difference of predicted and actual magnitude, has been calculated for each model. The prediction error obtained using Support Vector Machine (Radial basis function kernel), Lasso Regressor, Decision Tree Regressor and XGBoost model is obtained as 0.49, 0.65, 0.47 and 0.42, respectively, thereby indicating that the performance of the XGBoost model is best among these models. In the present work, we are calculating single integral, double integral and fraction, and our results may sometimes give out-of-bound results. XGBoost handles this case by pushing

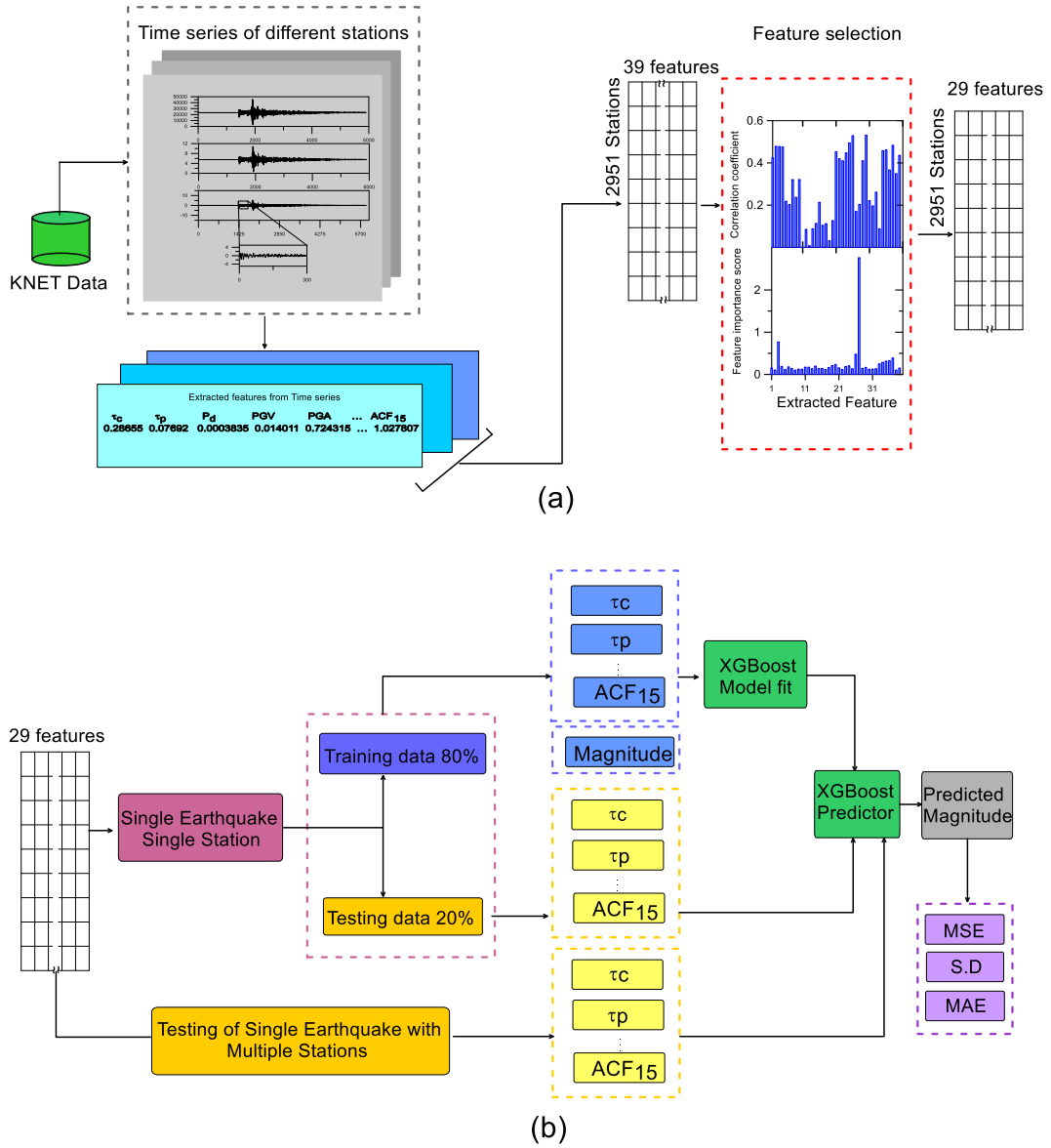


Figure 4. **(a)** Use of K-NET data for feature extraction and selection of features for XGBoost model and **(b)** use of selected features for training and testing of the XGBoost model for prediction of the earthquake magnitude.

these values to the right node if not mentioned by default. In contrast, other models do not handle this case.

The task of optimising the error produced by the machine learning model by tuning the input parameters is called hyperparameter optimisation, and the input parameters are called hyperparameters (Ørebæk and Geitle 2021). Hyperparameter tuning is a non-trivial process and can vary from input to input. The hyperparameter tuning of XGBoost model is done using grid search. Table 1 shows the final value obtained after tuning hyperparameters of XGBoost model. The XGBoost model grows the decision tree horizontally (levelwise) (Rezazadeh 2020) as shown in figure 5(a). Figure 5(a) represents the method of results by n estimators produced by the XGBoost model used in this paper. Shapley (SHAP) values for each feature has been calculated using the method introduced by Lundberg and Lee (2017) and is shown in figure 5(b and c), which represents the impact of various features on training of the model. It is seen from figure 5(b and c) that '*IV2*' feature is the most important feature, while ACF_{11} is the least one. The feature *IV2* has been referred to as square velocity integral and has already been used by Festa *et al.* (2008) for discriminating the magnitude of Japanese earthquake from the early P wave phase and has been reported as an estimator of the magnitude of an earthquake.

Table 1. *Training hyperparameters for gradient tree boosting technique using XGBoost algorithm.*

Parameter	Value	Description
max_depth	6	Maximum depth of the trees (Ma *et al.* 2020).
learning_rate	0.2	Step size used for optimising model (Wang and Ni 2019).
n_estimators	300	Number of estimators for boosting (Ma *et al.* 2020).
min_child_weight	2	Minimum instance weights that a leaf require (Wang and Ni 2019).
reg_alpha	0.005	L1 regularization term (Ma *et al.*, 2020)
reg_lambda	6	L2 regularization term (Feng *et al.* 2020)
gamma	0	Minimum loss reduction needed for further partitioning (Wang and Ni 2019).

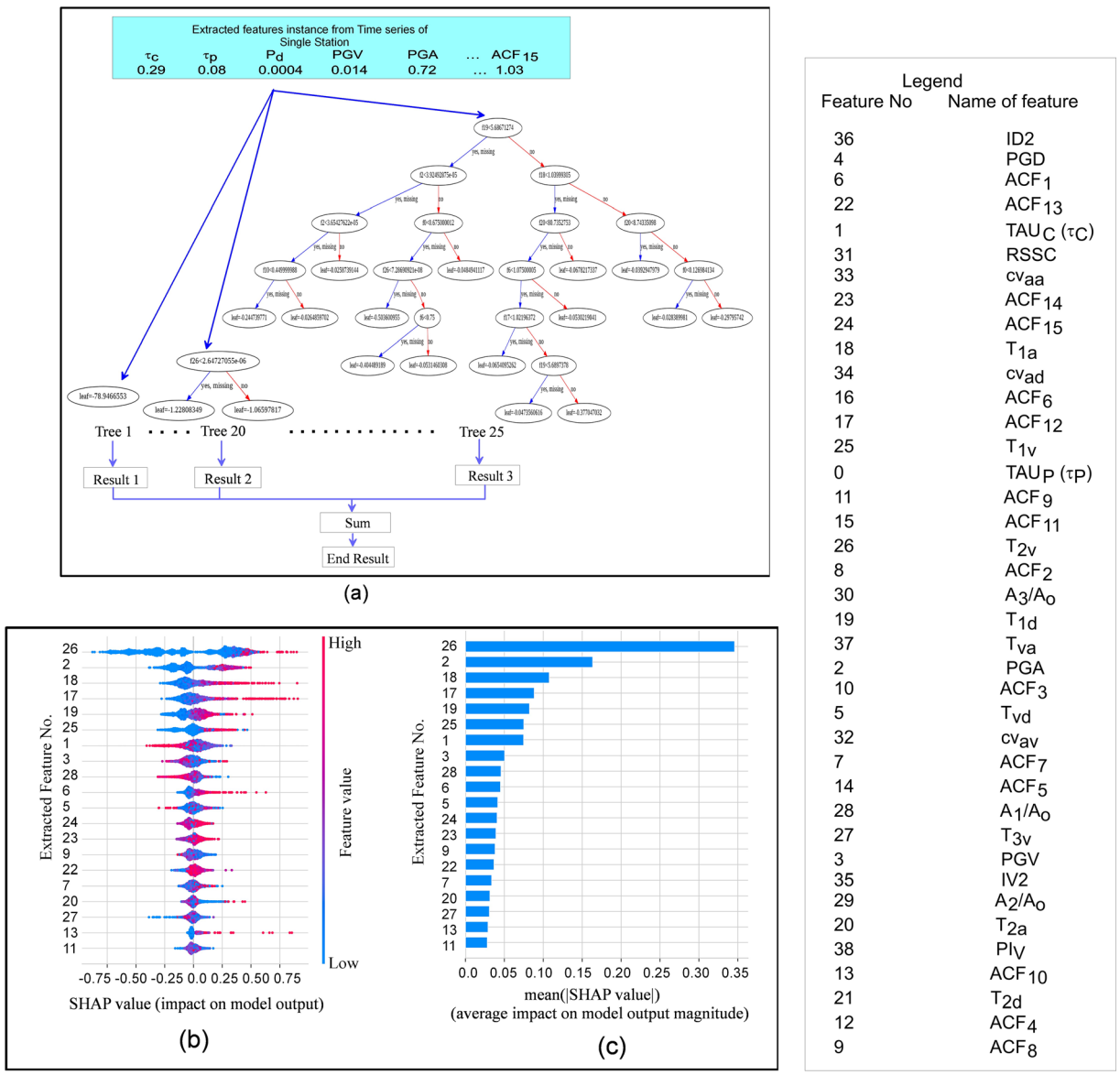


Figure 5. **(a)** Structure of XGBoost modified after Wang *et al.* (2020), **(b)** study of the impact of the extracted feature on the XGBoost model (blue for low and red for high), and **(c)** average impact on model prediction. Features 1–29 are related to extracted features defined in the legend of the figure.

3.1 *Feature engineering*

Once the entire records have been processed, the next task is to extract features from the early P phase of the records. There are only a few features extracted from the early part of the earthquake ground motions, possibility of many other parameters have also been explored, which can have a correlation with the magnitude of an earthquake. The parameters from ground acceleration have been extracted together with the converted velocity and displacement records and their auto-correlation functions. The acceleration time-series recorded by K-NET stations have been converted into velocity and displacement time series by using numerical integration schemes. After each integration process, low-cut filters have been applied to remove the undesired high-period noises in the records. A total of 39 features have been extracted from the early part of the acceleration, velocity and displacement records and their amplitude spectra and auto-correlation functions (ACF). Many parameters used in the process of feature extraction, like τ_c, τ_p and P_d, have been utilised in the EEW systems (Kanamori 2005; Wu and Kanamori 2005). Other parameters, mainly from the auto-correlation function, have already been applied in the oil industry for the discrimination of subsurface lithostratigraphy (Sinvhal and Khattri 1983) and have been utilised for magnitude prediction first time in the present work. Definitions of the various extracted features used in the training of the machine learning model have been defined in the following section.

Tauc (τ_c): This is the parameter which has been first given by Kanamori (2005) for the EEW system and is calculated from the vertical component of velocity '$v(t)$' and displacement '$d(t)$' record after the arrival of P phase by using the following formula:

$$r = \frac{\int_0^T |v^2(t)|dt}{\int_0^T |d^2(t)|dt}, \tag{5}$$

$$\tau_c = \frac{2\pi}{\sqrt{r}}. \tag{6}$$

Taup (τ_p): This is defined as the maximum predominant period observed in the velocity record after the onset of P phase. This parameter has been applied by Wu and Kanamori (2005) for the EEW system.

***PGD*, *PGA* and *PGV*:** *PGD* refer to the absolute peak displacement and can be computed from the displacement record. This parameter has been used by Wu and Kanamori (2005) for early warning approaches. In the present work, absolute peak ground acceleration (PGA) and peak ground velocity (PGV) are calculated from the record of the P phase.

***T_{1v}*, *T_{2v}*, *T_{3v}*, *T_{1d}*, *T_{2d}*, *T_{1a}*, *T_{2a}*:** These parameters represent the time of first and second zero crossing in the ACF. These parameters have been utilised as the main features by Sinvhal and Khattri (1983) for discriminating oil-bearing structures from seismic signals. In the present work, T_{1v}, T_{2v} and T_{3v} are first, second, and third zero-crossings and are computed from ACF of velocity records, while T_{1d}, T_{2d} and T_{1a}, T_{2a} are zero crossings computed from ACF of displacement and acceleration records, respectively.

***A1/A0*, *A2/A0*, *A3/A0*:** These parameters have been extracted from the amplitude of ACF function of velocity record of P phase. The parameters *A0*, *A1*, *A2*, and *A3* represent the value of ACF function at zero, one, two, and three lags, respectively. These parameters have already been extracted from seismic signals by Sinvhal and Khattri (1983) for discriminating oil-bearing structures.

***T_{va}* and *T_{vd}*:** The ratio of peak amplitude in acceleration and velocity records reflects the frequency components of the ground motion and correlates with magnitude (Böse 2006; Ma 2008). The parameter T_{va} is calculated as (Zhu *et al.* 2021):

$$T_{va} = 2\pi\left(\frac{PGV}{PGA}\right). \tag{7}$$

The parameter T_{vd} has been extracted in the present work from velocity and displacement records using the following relationship given by Zhu *et al.* (2021):

$$T_{vd} = \frac{PGV}{PGD}. \tag{8}$$

***PI_V*, *IV2*, *ID2*, *cv_{ad}*, *cv_{av}*, *cv_{aa}*:** Extracted parameters from the P phase, which are related to the magnitude of earthquakes, include the P-wave index value (PI_V) (Nakamura 2003) and velocity squared integral ($IV2$) (Festa *et al.* 2008). These parameters have been calculated as:

$$PI_V = 0 \leq t \leq T \log|a(t) \cdot v(t)| \tag{9}$$

$$IV2 = \int_0^T v^2(t)\,dt. \tag{10}$$

In the present work, $ID2$ has been calculated by substituting the displacement record in place of the velocity record in the definition of $IV2$. The parameter cv_{ad} represents cumulative vertical absolute displacement, cv_{av} represents cumulative vertical absolute velocity, and cv_{aa} represents cumulative vertical absolute acceleration. These parameters have been utilised by Zhu *et al.* (2021) to represent their dependency on the magnitude of an earthquake and are calculated as:

$$cv_{ad} = \sum_{0 \leq t \leq T} |d(t)|, \tag{11}$$

$$cv_{av} = \sum_{0 \leq t \leq T} |v(t)|, \tag{12}$$

$$cv_{aa} = \sum_{0 \leq t \leq T} |a(t)|. \tag{13}$$

***RSSC*:** This parameter has been extracted from the velocity record by the following formula:

$$RSSC = \sum \sqrt{v^2(t)}. \tag{14}$$

***ACF_1, ACF_2, ACF_3, ACF_4, ACF_5, ACF_6, ACF_7, ACF_8, ACF_9, ACF_{10}, ACF_{11}, ACF_{12}, ACF_{13}, ACF_{14} and ACF_{15}*:** These parameters have been extracted from the ACF of velocity, acceleration and displacement records of P phase. These parameters are calculated as:

ACF_1 = Time of first positive peak in ACF of acceleration record
ACF_2 = Time of second positive peak in ACF of acceleration record
ACF_3 = Time of first positive peak in ACF of velocity record
ACF_4 = Time of second positive peak in ACF of velocity record
ACF_5 = Time of first positive peak in ACF of displacement record
ACF_6 = Time of second positive peak in ACF of displacement record
ACF_7 = Time of first negative peak in ACF of acceleration record
ACF_8 = Time of second negative peak in ACF of acceleration record
ACF_9 = Time of first negative peak in ACF of velocity record
ACF_{10} = Time of second negative peak in ACF of velocity record
ACF_{11} = Time of first negative peak in ACF of displacement record
ACF_{12} = Time of second negative peak in ACF of displacement record
ACF_{13} = Ratio of area on the positive and negative side of the abscissa of the ACF function obtained from the acceleration records
ACF_{14} = Ratio of area on the positive and negative side of the abscissa of the ACF function obtained from the velocity records
ACF_{15} = Ratio of area on the positive and negative side of the abscissa of the ACF function obtained from the displacement records.

In this work, a Python program named FETS has been developed to process the acceleration data downloaded from K-NET site. The program extracts the mentioned features from the early 3 sec of P waveform. The program preprocesses the waveform by applying scaling and baseline correction and then extracts the above-mentioned magnitude-dependent parameters. The program FETS has been utilised to extract 39 features from the accelerogram. Once the features have been extracted from the record, their correlation with the magnitude of earthquake has been calculated using the Pearson correlation coefficient. The following formula for calculating the Pearson correlation coefficient between two variables, x_i and x_j, has been utilised in this work (Chen *et al.* 2019):

$$c = \frac{\sum_{l=1}^{N}(x_i - \overline{x}_i)(x_j - \overline{x}_j)}{\sqrt{\sum_{l=1}^{N}(x_i - \overline{x}_i)^2 \sum_{l=1}^{N}(x_j - \overline{x}_j)^2}}. \tag{15}$$

In this equation, $\overline{x}_i$ and $\overline{x}_j$ represent the average score of x_i and x_j features extracted from the time series of the entire dataset. Value of c lies in between +1 and –1, indicating a strong positive and negative correlation. The value of $|c| > 0.5$ indicates a strong correlation between the two features (Chen *et al.* 2019). The correlation matrix calculated using the above equation has been used to check the correlation of extracted features with the magnitude of earthquake. Figure 6 shows the correlation coefficients of various features with the magnitude of the earthquake and model-based feature importance scores calculated for different features. Other than

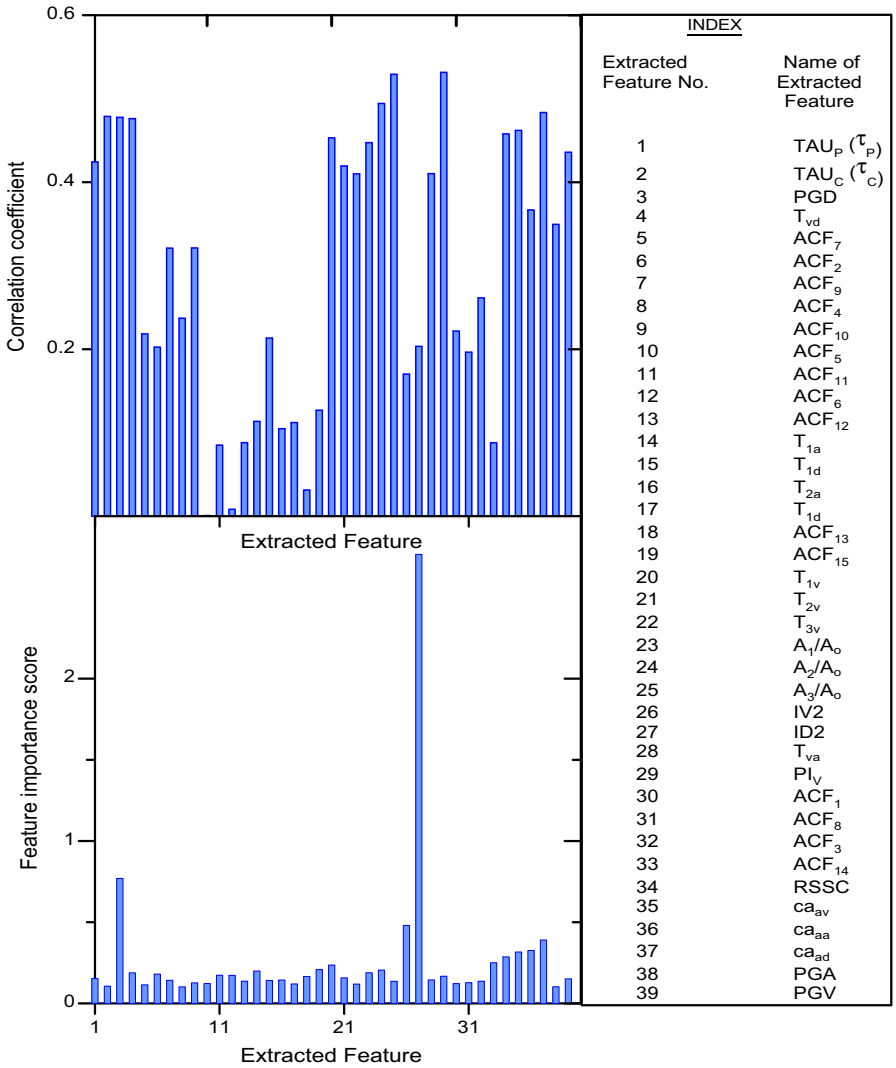


Figure 6. Correlation coefficient and model based feature importance score of 39 features extracted for the training of XGBoost model.

selecting redundant features on the basis of correlation coefficient value and model-based feature score, some features exhibiting average correlation and feature importance score but building unstable models and decelerating the training process are removed from the model's training process. A total of 29 features have been retained for training the machine learning model in this paper out of selected 39 features.

4. Results

After training of the XGBoost model by using P wave records of 3 sec from earthquakes of magnitude 3.0–7.4 recorded at different stations within 50–150 km hypocentral distance, the system has been passed through test data, which is completely independent of the training data. A total of 591 records randomly selected from the entire data have been used for testing the trained XGBoost model. The τ_c and P_d methods are widely used in the study of EEW systems for magnitude prediction (Kanamori 2005; Wu and Kanamori 2005; Wu and Li 2006; Zollo *et al.* 2006; Colombelli *et al.* 2014). The performance of the trained XGBoost model has been evaluated by comparing the magnitude predicted by using τ_c and P_d methods. The magnitude has been calculated using τ_c and P_d parameters from early 3 sec of P phase using the following relations given by Jin *et al.* (2013) for Japanese earthquake:

$$M = 2.94\log_{10}(\tau_c) + 5.3 \pm 0.46, \tag{16}$$

$$M = 0.91\log_{10}(P_d) + 0.48\log_{10}(hy) + 5.65 \pm 0.56. \tag{17}$$

In the above expression, parameter '*hy*' denotes the hypocentral distance in km. It is seen from

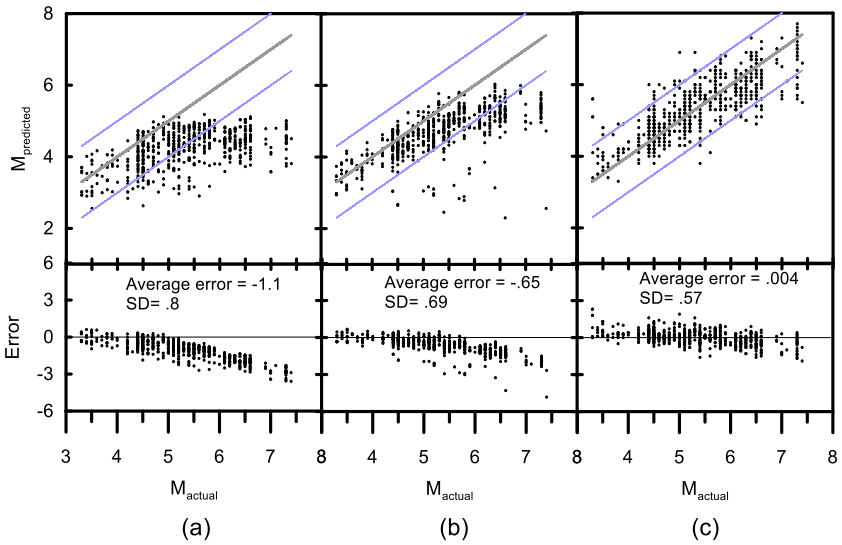


Figure 7. Distribution of predicted magnitude and prediction error with respect to actual magnitude using **(a)** τ_c method, **(b)** P_d method, and **(c)** XGBoost model. The two blue solid lines indicate the locations of one standard deviation of error. The grey line serves as a guideline indicating that the data points are expected to fall along it.

figure 7 that the predicted earthquake magnitude calculated from the early 3 sec of P wave record using XGboost model gives average prediction error, which is minimum among those obtained using τ_c and P_d methods. The 591 records of different earthquakes are passed through XGBoost model to predict the magnitude of earthquake. The average error is obtained as –1.1, –0.65 and 0.004 for predictions using τ_c, P_d and the XGBoost models, while the standard deviation obtained in the prediction error is 0.80, 0.69 and 0.57, respectively, for predictions using τ_c, P_d and the XGBoost model. The comparison of prediction error with actual magnitude for three different cases clearly shows that error increases with an increase in magnitude for both τ_c and P_d methods, while it shows scattered values with respect to zero prediction error for the XGBoost model. The magnitude predicted using τ_c and P_d methods gives underpredicted values that increase in the higher magnitude range, which indicates that predicted magnitude is highly dependent on the magnitude of earthquake and possibility of underprediction is high for high-magnitude earthquakes. It is seen that the average prediction error obtained using XGBoost method is nearly zero, which indicates that magnitude determined using XGBoost model remains unaffected by the size of an earthquake. The comparison of predicted and actual magnitude in figure 7 indicates that τ_c method gives saturated values at 5.0 magnitude, which is further enhanced for P_d method up to 6.0 magnitude. Further, the trained XGBoost model shows signs of saturation at a higher magnitude of 7.3, which is higher among these three methods.

The distribution of prediction error with respect to the hypocentral distance shown in figure 8(a, b and c) indicates that both conventional τ_c and P_d methods give a high value of prediction error on average, which increases with hypocentral distance, while very low prediction error is obtained in case of XGBoost model. It is seen that a high amount of discreteness is exhibited in the prediction error obtained by the τ_c and P_d methods, and the average prediction error clearly varies with increasing hypocentral distance. The average prediction error using the XGBoost model remains nearly equal to zero for all hypocentral distances, which indicates that magnitude estimation errors are unaffected by the choice of hypocentral distance considered in the XGBoost model. The response of τ_c, P_d and the XGBoost model in the prediction of earthquake magnitude from records in the close hypocentral distance range, i.e., 50–75 km, have been shown in figure 8(d, e and f), respectively. It is seen that prediction error increases with distance in τ_c and P_d methods and exhibits more discreteness compared to the XGBoost model. Further, it is seen that the average prediction error in the XGboost model remains nearly equal to zero, thereby indicating that the trained model is working properly in the near-source regions, which are areas of major destruction during an earthquake.

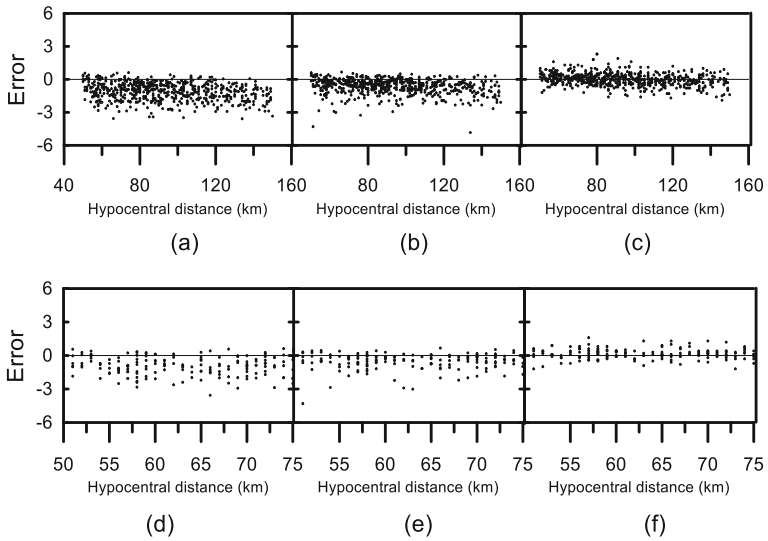


Figure 8. Distribution of prediction error with respect to the hypocentral distance for **(a)** τ_c method, **(b)** P_d method, and **(c)** XGBoost model. Distribution of prediction error computed using records from station within hypocentral distance of 50–75 km for **(d)** τ_c method, **(e)** P_d method, and **(f)** XGBoost model.

In order to have a closer look at the response of a trained XGBoost model for an actual scenario, two earthquakes of magnitude 4.5 and 6.1 (M_{JMA}) are selected for multi-station magnitude prediction study. These magnitudes have been recorded at 26 and 30 stations lying in a hypocentral distance of 50–150 km that has been passed through the XGBoost model. Location of the epicentre of these earthquakes, together with the recording stations that have been used in this paper, are shown in figures 9(a) and 10(a). Vertical component of the acceleration time series recorded at some of the stations is shown in figures 9(b) and 10(b). The parameters of these earthquakes are given in table 2. The magnitude predicted by the XGBoost model using different input records for 4.5 M_{JMA} has been compared with that calculated using the τ_c and P_d methods using regression relations given in equations (16 and 17) and is shown in figure 9(c, d and e), respectively. The average predicted magnitude obtained from the first 3 sec of record after the arrival of P phase is 4.37 ± 0.35 in the case of the XGBoost model, while it is 3.58 ± 0.34 and 4.02 ± 0.33 in the case of τ_c and P_d methods. Similarly, the performance of the model for the 6.1 M_{JMA} with respect to the τ_c and P_d methods using regression relations has been shown in figure 10(c, d, and e). The average magnitude predicted and the standard deviation obtained from the selected 6.1 M_{JMA} earthquake for the multi-station earthquake analysis study is 6.17 ± 0.30 in the case of the XGBoost model, while it is 3.98 ± 0.32 and 4.82 ± 0.29 in the case of τ_c and P_d methods. It is seen that both τ_c and P_d methods are underestimating the magnitude while the prediction from XGBoost model is close to actual observation. Further, it is seen that the predicted magnitude from those stations lying within a hypocentral distance of 50–70 km gives an average predicted magnitude of 4.58 ± 0.33 and 6.32 ± 0.29, which is close to the actual magnitude of 4.5 and 6.1 M_{JMA} of the earthquake and are fair enough to give warning to other stations before the arrival of seismic energy at these stations. The comparison of the prediction error from the trained XGBoost model with that obtained from τ_c and P_d methods clearly establishes the efficacy of the XGBoost model and its utility in the EEW system.

5. Discussion

Several numerical tests have been performed before finalising the training dataset and hyperparameters of XGBoost. The model has been trained using the gradient tree boosting technique of XGBoost from the 29 extracted features from 2360 records. The average (μ) and standard deviation (σ) of prediction error data have been calculated using the following formula:

$$\mu = \frac{\sum\left(M_{\text{pred}} - M_{\text{actual}}\right)}{n}, \tag{18}$$

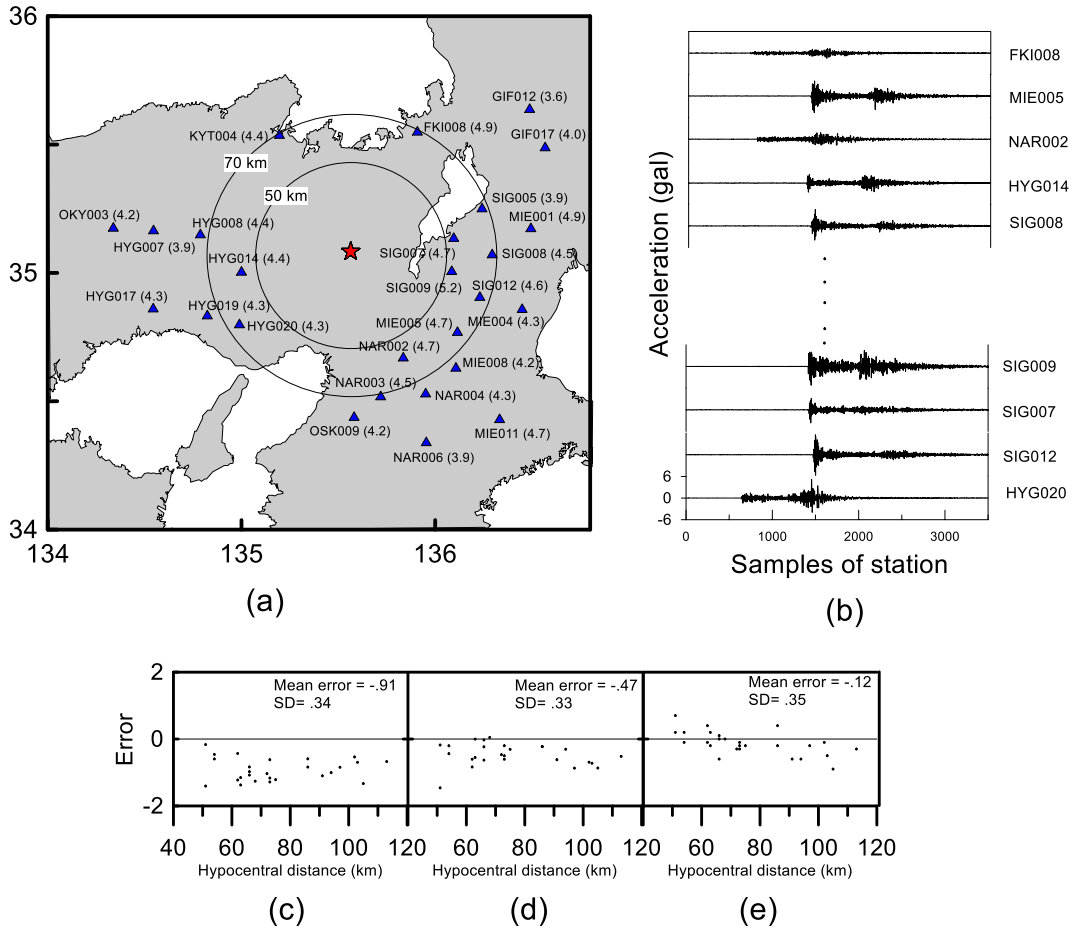


Figure 9. **(a)** Location of the epicentre (star) of the 21/12/2007 earthquake and stations (triangle) that have been used as testing data for predicting magnitude using XGBoost model. Estimated magnitude by XGBoost model at each station is shown by number along with the station code. **(b)** Acceleration time series of some of the stations that have recorded this earthquake. Predicted error in magnitude calculation using **(c)** τ_c, **(d)** P_d, and **(e)** XGBoost model.

$$\sigma = \sqrt{\frac{\sum\left(\left|M_{\text{pred}} - M_{\text{actual}}\right|^2\right)}{n}}. \tag{19}$$

In the above equation, M_{pred} and M_{actual} denote the predicted and actual magnitude, respectively. Although absolute prediction error can also give an overall idea of prediction error, the positive and negative value of average prediction error gives an idea of over- and under-predicted cases reported by the model.

The trained XGBoost model has been tested for predicting magnitudes from the extracted features of records from test data that are independent of the training data. Magnitude has been predicted from trained models after extracting 29 features from records of 2.0, 2.5, 2.8 and 3.0 sec duration from the onset of the P phase. It is seen from figure 11 that the obtained average and standard deviation of prediction error is minimum for the case when 3.0 seconds of records have been utilised in the training of the model. Therefore, in this work, 3.0 sec of record after the onset of P phase has been utilised for training purposes. It has been demonstrated by Wu and Li (2006) and Wu *et al.* (2007) that a record of 3 sec after the onset of P phase is suitable for advanced prediction of earthquake magnitude.

Dependency of the XGBoost models on the range of input magnitude used in the training dataset has been numerically tested by considering different training datasets that have minimum cut-off magnitude of 3.0, 3.5, 4.0, 4.5, 5.0, 5.5 and 6.0, respectively. The number of records used for training of XGBoost models by using datasets having minimum cut-off magnitudes of 3.0, 3.5, 4.0, 4.5, 5.0, 5.5 and 6.0 is 2360, 2325, 2184, 1878, 1360, 864, 502, respectively. These trained models are tested with independent test datasets having a similar cut-off magnitude range. The training and testing datasets are independent and are shown in figure 12(a). It is seen that the prediction error is

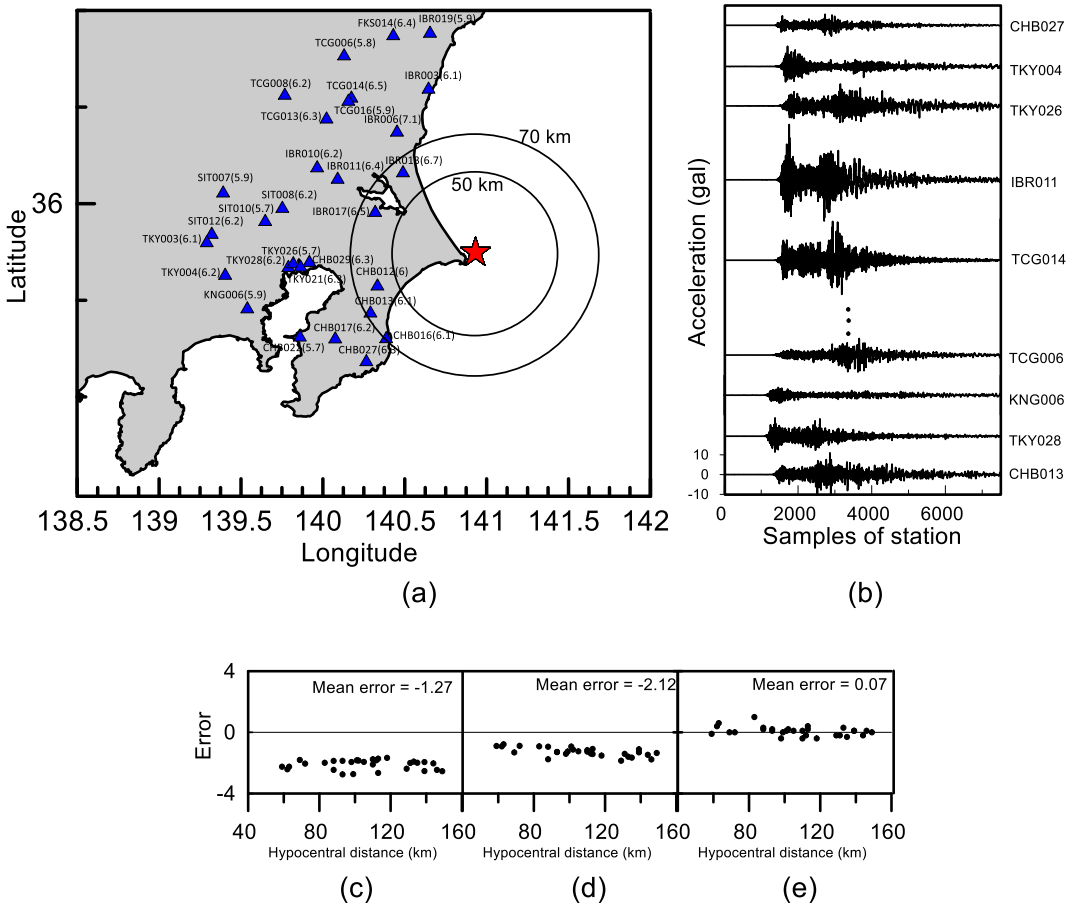


Figure 10. **(a)** Location of epicentre (star) of 21/12/2007 earthquake and stations (triangle) that have been used as testing data for predicting magnitude using XGBoost model. **(b)** The raw acceleration records of the selected stations for this earthquake. The error prediction in magnitude calculation using **(c)** τ_c, **(d)** P_d, and **(e)** XGBoost model.

Table 2. *Parameters of earthquakes used as test case.*

Parameter	Case 1	Case 2
Origin	21/12/2007, 14:23:00	14/03/2012, 21:05:00
Epicentre	35.083, 135.564	35.747, 140.932
Magnitude	4.5 (M_{JMA})	6.1 (M_{JMA})
Hypocenter depth (km)	7	15

minimum for training datasets having magnitude range from 3.0 to 7.4 and increases sharply with an increase in the minimum cut-off magnitude range, as shown in figure 12(a). It is clearly seen that the prediction by trained Xboost model is more reliable when the training data is complete and the prediction error increases with decrease in the quantity of the good dataset.

Hypocentral distance is an important parameter that governs the amount of destruction during an earthquake. Primary aim of any EEW network is to predict earthquake magnitude from records in the near-source region, i.e., at a very low hypocentral distance. However, for such records, the difference in the arrival time of P and S wave is very less, and hence, there is always the possibility of overlapping the initial S phase with the prominent P phase. This may result in over- and under-predictions of magnitude. A numerical test has been performed for training and testing of XGBoost model through a database, which differs in terms of hypocentral distance. Nine different training sets have been prepared in which minimum hypocentral distance (cut-off hypocentral distance) is assumed as 50, 60, 70, 80, 100, 110, 120 and 130 km, respectively. In these training sets, the

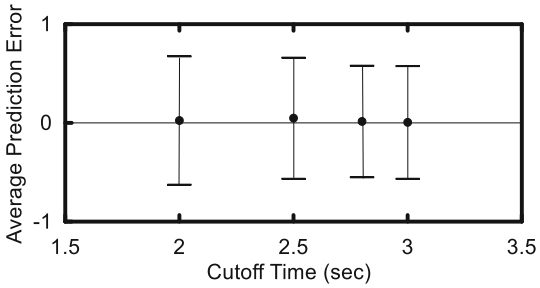


Figure 11. Average and standard deviation of prediction error obtained using XGBoost models which records up to 2, 2.5, 2.8 and 3.0 sec, respectively after the onset of P phase. The vertical line indicates standard deviation in the prediction error.

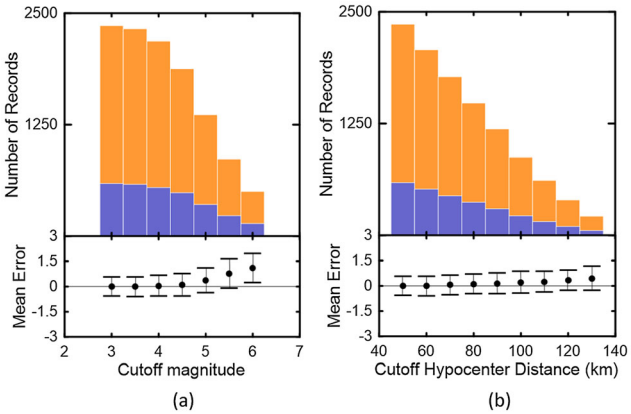


Figure 12. **(a)** Average magnitude prediction error in the test data obtained from the XGBoost model prepared using the training set by considering training data having a cut-off magnitude of 3.0, 3.5, 4.0, 4.5, 5.0, 5.5 and 6.0, respectively. Orange and blue bars show the training and test data, respectively. **(b)** Average magnitude prediction error in the test data obtained from the XGBoost model prepared using the training set by considering training data having a cut-off hypocentral distance of 50, 60, 70, 80, 100, 110, 120 and 130 km, respectively. Orange and blue bars show the training and test data, respectively.

maximum limit of the hypocentral distance is kept as 150 km. Different test data shown in figure 12(b) have been passed through nine different trained models and prediction error has been calculated and is shown in figure 12(b). It is seen that prediction errors are minimum when the dataset covers hypocentral distance of more than 50 km.

The dependnency of XGBoost model on magnitude and hypocentral distance is explained on account of data used in the training of model. It is seen that higher cut-off magnitude in the dataset results in lesser number of records used for training the model, which ultimately results in higher prediction error. Similarly, increase in the cut-off hypocentral distance indicates lesser number of data used for training the model. This has been clearly explained in figure 12, which indicate that the XGBoost model trained using complete dataset that covers records in a wider range of magnitude and hypocentral distance is most suitable for prediction. The trained XGBoost model has been tested with the dataset having similar range of data used by one of the recent machine-learning model proposed by Zhu *et al.* (2022). The machine learning model by Zhu *et al.* (2022) utilised the dataset in a magnitude range from 3 to 6.5 (M_{JMA}) with a mean absolute error close to 0.5. The mean absolute error obtained by the XGBoost model for similar data range as used by Zhu *et al.* (2022) is obtained as 0.39, thereby confirming the efficacy of the developed model for the prediction of earthquake magnitude.

6. Conclusions

In the present work, the XGBoost model based on gradient tree boosting techniques has been utilised for the prediction of earthquake magnitude from early P phase of the record. The earthquakes recorded by K-NET that have occurred in the part of Japan Islands, which lie in the Eurasian Plate, have been utilised for training and testing this model. Several numerical tests have been performed for finalising the range of magnitude and hypocentral distance in the testing and training datasets. The reliability of the trained model has been tested on the basis of its ability to correctly predict the magnitude of an earthquake using records from single or multiple stations in terms of prediction error. The predicted magnitude from the trained XGBoost model is compared with that obtained from the same test dataset using τ_c and P_d parameters. The trained XGBoost model gives an average prediction error of 0.004 ± 0.57 for all input records in the test dataset, while the average prediction error is obtained as -1.1 ± 0.80 and -0.65 ± 0.69 for the same dataset when estimating magnitude by using τ_c and P_d methods, respectively.

In order to have a closer look at the response of a trained system for an actual scenario, 26 and 30 records of an earthquake of magnitude 4.5 and 6.1 M_{JMA} that are independent of the training and testing dataset have been used as input to the trained XGBoost model. It is seen that the model gives an average predicted magnitude of 4.58 ± 0.33 and 6.32 ± 0.29 from early P phase of

nearfield records, which is close to the actual magnitude and is fair enough to give warning to other stations before the arrival of seismic energy at other stations. Comparison of magnitude prediction error by the existing conventional methods with that obtained from XGBoost model clearly establishes the efficacy of the trained model presented in this paper and its utility in the EEW system. The results presented in this paper have shown that the early part of earthquake ground motion can be effectively used to predict the magnitude of earthquakes using a machine learning model with a fair amount of accuracy.

Acknowledgements

Authors sincerely thank the National Research Institute for Earth Science and Disaster Resilience (2019) for providing K-NET station strong motion data used in this work. The research done in this paper is the outcome of the sponsored research project grant no. MoES/P/O.(Seismo)/I(397)/2021. Data has been obtained from the National Research Institute for Earth Science and Disaster Resilience (2019).

Author statement

Anushka Joshi: Developed code for processing, feature extraction, training and testing of XGBoost model and figure development. Chalavadi Vishnu: Literature survey, figure development, manuscript preparation, and flow graph of process. C Krishna Mohan: Interpretation of results and mathematical formulation.

References

Alexandridis A, Chondrodima E, Efthimiou E, Papadakis G, Vallianatos F and Triantis D 2014 Large earthquake occurrence estimation based on radial basis function neural networks; *IEEE Trans. Geosci. Remote Sens.* **52** 5443–5453.

Allen R V 1978 Automatic earthquake recognition and timing from single traces; *Bull. Seismol. Soc. Am.* **68** 1521–1532, https://doi.org/10.1007/BF02247958.

Allen R M and Kanamori H 2003 The potential for earthquake early warning in southern California; *Science* **300** 786–789.

Allen R M, Gasparini P, Kamigaichi O and Böse M 2009 The status of earthquake early warning around the world: An introductory overview; *Seismol. Res. Lett.* **80** 682–693.

Alves E I 2006 Earthquake forecasting using neural networks: Results and future work; *Nonlinear Dyn.* **44** 341–349.

Aoi S, Kunugi T, Nakamura H and Fujiwara H 2011 Deployment of new strong-motion seismographs of K-NET and KiK-net; *Earthq. Data Eng. Seismol. Geotech. Geol. Earthq. Eng.* **14** 167–186, https://doi.org/10.1007/978-94-007-0152-6_12.

Aoi S, Asano Y, Kunugi T, Kimura T, Uehira K, Takahashi N, Ueda H, Shiomi K, Matsumoto T and Fujiwara H 2020 MOWLAS: NIED observation network for earthquake, tsunami and volcano; *Earth Planet. Space* **72** 1–31.

Båth M 1952 Earthquake magnitude determination from the vertical component of surface waves; *Eos Trans. Am. Geophys. Union* **33** 81–90.

Bekkerman R, Bilenko M and Langford J 2011 *Scaling up machine learning: Parallel and distributed approaches*; Cambridge University Press, New York, NY, USA.

Boore D M and Bommer J J 2005 Processing of strong-motion accelerograms: Needs, options and consequences; *Soil. Dyn. Earthq. Eng.* **25** 93–115.

Böse M 2006 Earthquake early warning for Istanbul using artificial neural networks; *PhD thesis*, University of Karlsruhe, Karlsruhe.

Böse M, Hauksson E, Solanki K, Kanamori H, Wu Y M and Heaton T H 2009 A new trigger criterion for improved real-time performance of onsite earthquake early warning in southern California; *Bull. Seismol. Soc. Am.* **99** 897–905.

Böse M, Ionescu C and Wenzel F 2007 Earthquake early warning for Bucharest, Romania: Novel and revised scaling relations; *Geophys. Res. Lett.* **34**, https://doi.org/10.1029/2007GL029396.

Brown H, Allen M, Richard M and Grasso V F 2009 Testing ElarmS in Japan; *Seismol. Res. Lett.* **80** 727–739.

Chamoli B P, Kumar A, Chen D, Gairola A, Jakka R S, Pandey B, Kumar P and Rathore G 2021 A prototype earthquake early warning system for northern India; *J. Earthq. Eng.* **25** 2455–2473.

Chen T and Guestrin C 2016 XGBoost: A scalable tree boosting system; *KDD '16: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Chen M, Liu Q, Chen S, Liu Y, Zhang C and Liu R 2019 XGBoost-based algorithm interpretation and application on post-fault transient stability status prediction of power system; *IEEE Access.* **7** 13,149–13,158.

Colombelli S, Zollo A, Festa G and Picozzi M 2014 Evidence for a difference in rupture initiation between small and large earthquakes; *Nat. Comm.* **5** 3958.

Cua G, Fischer M, Heaton T and Wiemer S 2009 Real-time performance of the virtual seismologist earthquake early warning algorithm in southern California; *Seismol. Res. Lett.* **80** 740–747.

Cuéllar A, Espinosa-Aranda J M, Suárez G, Ibarrola G, Uribe A, Rodríguez F H, Islas R, Rodríguez G M, García A and Frontana B 2014 The Mexican Seismic Alert System (SASMEX): Its alert signals, broadcast results and performance during the M 7.4 Punta Maldonado Earthquake of March 20th, 2012; In: *Early warning for geological disasters: Scientific methods and current practice* (eds) Wenzel F and Zschau J, Springer, Berlin, Heidelberg, pp. 71–87.

Elgamal A and He L 2004 Vertical earthquake ground motion records: An overview; *J. Earthq. Eng.* **8** 663–697.

Feng Y, Wang D and Yin Y 2020 An XGBoost-based casualty prediction method for terrorist attacks; *Complex Intell. Syst.* **6** 721–740.

Festa G, Zollo A and Lancieri M 2008 Earthquake magnitude estimation from early radiated energy; *Geophys. Res. Lett.* **35** L22307, https://doi.org/10.1029/2008gl035576.

Friedman J H 2001 Greedy function approximation: A gradient boosting machine; *Ann. Statist.* **29** 1189–1232.

Given D, Cochran E, Heaton T, Hauksson E, Allen R, Hellweg M, Vidale J and Bodin P 2014 Technical implementation plan for the ShakeAlert production prototype system – an earthquake early warning system for the west coast of the United States; *U.S. Geological Survey Open-File Report*, https://doi.org/10.3133/ofr20141097.

Grigoli F, Scarabello L, Böse M, Weber B, Wiemer S and Clinton J F 2018 Pick- and waveform-based techniques for real-time detection of induced seismicity; *Geophys. J. Int.* **213** 868–884.

Hoshiba M 2014 Review of the nationwide earthquake early warning in Japan during its first five years; In: *Earthquake hazards, risk and disasters* (ed.) Shroder J F, Academic Press, pp. 505–529.

Jin X, Zhang H, Li J, Wei Y and Ma Q 2013 Earthquake magnitude estimation using the τ_c and P_d method for earthquake early warning systems; *Earthq. Sci.* **26** 23–31.

Johnson, P A, Rouet-Leduc B, Pyrak-Nolte L J, Beroza G C, Marone C J, Hulbert C, Howard A, Singer P, Gordeev D, Karaflos D, Levinson C J, Pfeiffer P, Puk K M and Reade W 2021 Laboratory earthquake forecasting: A machine learning competition; *Proc. Nat. Acad. Sci. USA* **118(5)**.

Kanamori H 2005 Real-time seismology and earthquake damage mitigation; *Ann. Rev. Earth Planet. Sci.* **33** 195–214.

Kuyuk H and Allen R 2013 A global approach to provide magnitude estimates for earthquake early warning alerts; *Geophys. Res. Lett.* **40** 6329–6333.

Lundberg S M and Lee S 2017 A unified approach to interpreting model predictions; *Advances in Neural Information Processing Systems* **30** 4768–4777.

Ma Q 2008 Study and application on earthquake early warning; *PhD thesis,* Institute of Engineering Mechanics, Harbin.

Ma X, Fang C and Ji J 2020 Prediction of outdoor air temperature and humidity using XGBoost; *IOP Conf. Ser.: Earth Environ.* **427** 012013.

Mousavi S M and Beroza G C 2020 A machine-learning approach for earthquake magnitude estimation; *Geophys. Res. Lett.* **47**, https://doi.org/10.1029/2019gl085976.

Nakamura Y 1988 On the urgent earthquake detection and alarm system (UrEDAS); In: *Proc. 9th World Conf. Earthq. Eng.* **7** 673–678.

Nakamura Y 1989 Earthquake alarm system for Japan railways; *Jp. Railway Eng.* **28** 3–7.

Nakamura Y 2003 A new concept for the earthquake vulnerability estimation and its application to the early warning system; In: *Early warning systems for natural disaster reduction* (eds) Zschau J and Küppers A, Springer-Verlag, Berlin, pp. 693–699.

National Research Institute for Earth Science and Disaster Resilience 2019 NIED K-NET, KiK-net, National Research Institute for Earth Science and Disaster Resilience, https://doi.org/10.17598/NIED.0004.

Ochoa L H, Niño L F and Vargas C A 2017 Fast magnitude determination using a single seismological station record implementing machine learning techniques; *Geodes. Geodyn.* **9** 34–41.

Olivieri M, Allen R M and Wurman G 2008 The potential for earthquake early warning in Italy using ElarmS; *Bull. Seismol. Soc. Am.* **98** 495–503.

Ørebæk O and Geitle M 2021 Exploring the hyperparameters of XGBoost through 3D visualizations; In: *Proceedings of the AAAI 2021 Spring Symposium on Combining Machine Learning and Knowledge Engineering (AAAI-MAKE 2021)* (eds) Hinkelmann A M K, Fill H G, Gerber A, Lenat D, Stolle R and Harmelen F V, Stanford University, Palo Alto, California, USA.

Papazoglou A and Elnashai A S 1996 Analytical and field evidence of the damaging effect of vertical earthquake ground motion; *Earthq. Eng. Struct. Dyn.* **25** 1109–1137.

Perol T, Gharbi M and Denolle M 2018 Convolutional neural network for earthquake detection and location; *Sci. Adv.* **4** e1700578.

Peterson J 1993 Observations and modeling of seismic background noise; *US Geol. Surv. Open-File Rep.* **93-322**.

Picozzi M, Zollo A, Brondi P, Colombelli S, Elia L and Martino C 2015 Exploring the feasibility of a nationwide earthquake early warning system in Italy; *J. Geophys. Res.: Solid Earth* **120** 2446–2465.

Reddy R and Nair R R 2013 The efficacy of support vector machines (SVM) in robust determination of earthquake early warning magnitudes in central Japan; *J. Earth Syst. Sci.* **122** 1423–1434.

Rezazadeh A 2020 A generalised flow for B2B sales predictive modeling: An Azure machine-learning approach; *Forecasting* **2** 267–283.

Rouet-Leduc B, Hulbert C, Lubbers N, Barros K, Humphreys C J and Johnson P A 2017 Machine learning predicts laboratory earthquakes; *Geophys. Res. Lett.* **44** 9276–9282.

Satake K 2015 Geological and historical evidences of irregular recurrent earthquakes in Japan; *Phil. Trans. Roy. Soc. A: Math. Phys. Eng. Sci.* **373** 20140375.

Satriano C, Wu Y, Zollo A and Kanamori H 2011 Earthquake early warning: Concepts, methods and physical grounds; *Soil Dyn. Earthq. Eng.* **31** 106–118.

Shieh J, Wu Y and Allen R M 2008 A comparison of τ_c and τ_{pmax} for magnitude estimation in earthquake early warning; *Geophys. Res. Lett.* **35**, https://doi.org/10.1029/2008GL035611.

Sinvhal A and Khattri K N 1983 Application of seismic reflection data to discriminate subsurface lithostratigraphy; *Geophysics* **48** 498–1513.

Takla E M, Yumoto K, Okano S, Uozumi T and Abe S 2013 The signature of the 2011 Tohoku mega earthquake on the geomagnetic field measurements in Japan; *NRIAG J. Astron. Geophys.* **2** 85–195.

Wang Y and Ni X 2019 A XGBoost risk model via feature selection and Bayesian hyper-parameter optimisation; *ArXiv Abs*, https://doi.org/10.5121/ijdms.2019.11101.

Wang W, Chakraborty G and Chakraborty B 2020 Predicting the risk of chronic kidney disease (CKD) using machine learning algorithm; *Appl. Sci.* **11** 202.

Wenzel F, Oncescu M, Baur M and Fiedrich F 1999 An early warning system for Bucharest; *Seismol. Res. Lett.* **70** 161–169.

Wu Y M and Kanamori H 2005 Experiment on an onsite early warning method for the Taiwan early warning system; *Bull. Seismol. Soc. Am.* **95** 347–353.

Wu Y and Zhao Li 2006 Magnitude estimation using the first three second P-wave amplitude in earthquake early warning; *Geophys. Res. Lett.* **331**, https://doi.org/10.1029/2006GL026871.

Wu Y M, Kanamori H, Richard M A and Hauksson E 2007 Determination of earthquake early warning parameters, τ_c and P_d, for southern California; *Geophys. J. Int.* **170** 711–717.

Wu Y M, Hsiao N C, Chin T L, Chen D Y, Chan Y T and Wang K S 2014 Earthquake early warning system in Taiwan; In: *Earthquake engineering and structural dynamics* (eds) Beer M, Kougioumtzoglou I A, Patelli E and Au S-K, Springer, Berlin.

Yamamoto S, Rydelek P, Horiuchi S, Wu C and Nakamura H 2008 On the estimation of seismic intensity in earthquake early warning systems*; Geophys. Res. Lett.* **35,** https://doi.org/10.1029/2007GL033034.

Yamazaki F 1996 Earthquake monitoring and real-time damage assessment systems in Japan – developments and future directions; *Proc. 6th U.S.–Japan Workshop on Earthquake Resistant Design of Lifeline Facilities and Countermeasures Against Soil Liquefaction;* Technical Report NCEER-960012, pp. 727–740.

Yamazaki F, Noda S and Meguro K 1998 Developments of early earthquake damage assessment systems in Japan; *Proc. ICOSSAR'97, Structural Safety and Reliability*, pp. 1573–1580.

Zhang M, Ellsworth W L and Beroza G C 2019 Rapid earthquake association and location; *Seismol. Res. Lett.* **90** 2276–2284.

Zhang X, Zhang M and Tian X 2021 Real-time earthquake early warning with deep learning: Application to the 2016 M 6.0 Central Apennines, Italy Earthquake; *Geophys. Res. Lett.* **48**, https://doi.org/10.1029/2020GL089394.

Zhu J, Li S, Song J and Wang Y 2021 Magnitude estimation for earthquake early warning using a deep convolutional neural network; *Front. Earth Sci.* **9** 341.

Zhu J, Li S and Song J 2022 Magnitude estimation for earthquake early warning with multiple parameter inputs and a support vector machine; *Seismol. Res. Lett.* **93** 126–136.

Zollo A, Lancieri M and Nielsen S 2006 Earthquake magnitude estimation from peak amplitudes of very early seismic signals on strong motion records; *Geophys. Res. Lett.* **33** L23312, https://doi.org/10.1029/2006GL027795.

Corresponding editor: Sagarika Mukhopadhyay